本节内容

以太网交换机

408考研大纲（链路层部分）

（一）数据链路层的功能

（二）组帧

（三）差错控制

检错编码；纠错编码

（四）流量控制与可靠传输机制

流量控制、可靠传输与滑动窗口机制；停止-等待协议

后退 *N* 帧协议（GBN）；选择重传协议（SR）

（五）介质访问控制

1．信道划分：频分多路复用、时分多路复用、波分多路复用、码分多路复用

2．随机访问：ALOHA 协议；CSMA 协议；CSMA/CD 协议；CSMA/CA 协议

3．轮询访问：令牌传递协议

（六）局域网

局域网的基本概念与体系结构；以太网与 IEEE 802.3

IEEE 802.11 无线局域网；VLAN 基本概念与基本原理

（七）广域网

广域网的基本概念；PPP 协议

（八）数据链路层设备

以太网交换机及其工作原理

网桥（大纲已删）

交换机=多端口网桥

- 以太网交换机
  - 特点
    - 交换机=多端口***网桥***（408大纲已删除网桥）
    - 交换机工作在数据链路层，可以根据目的MAC地址转发帧
  - 自学习功能（支持即插即用）
    - 交换表 — 初始为空，记录【MAC地址，端口号】的对应关系
    - 每**收到一个帧**，就将**"发送方"**的【MAC地址，端口号】更新到交换表
    - ①如果**不知道"接收方"在哪里**，就把帧**广播到除入口外的其他端口**
      ②如果**知道"接收方"在哪里**，就把帧**精准转发**至某个端口
    - 交换表中每个表项都有"有效时间"，过期表项自动作废 — 以防某些节点拔线跑路
  - 两种交换方式
    - 直通交换
      - 只检查帧的目的MAC地址，以决定帧的转发端口
      - 优点：转发时延低
      - 缺点：不适用于需要速率匹配、协议转换或差错检测的线路
    - 存储转发交换
      - 先把帧完整地接收到交换机内部的高速缓存中，进行差错检测等必要处理，再根据交换表决定从哪个端口转发出去
      - 优点：适用于需要速率匹配、协议转换或差错检测的线路
      - 缺点：转发时延高

V2标准的以太网MAC帧：
6 6 2 N 4，收发协数验

以太网交换机的自学习功能

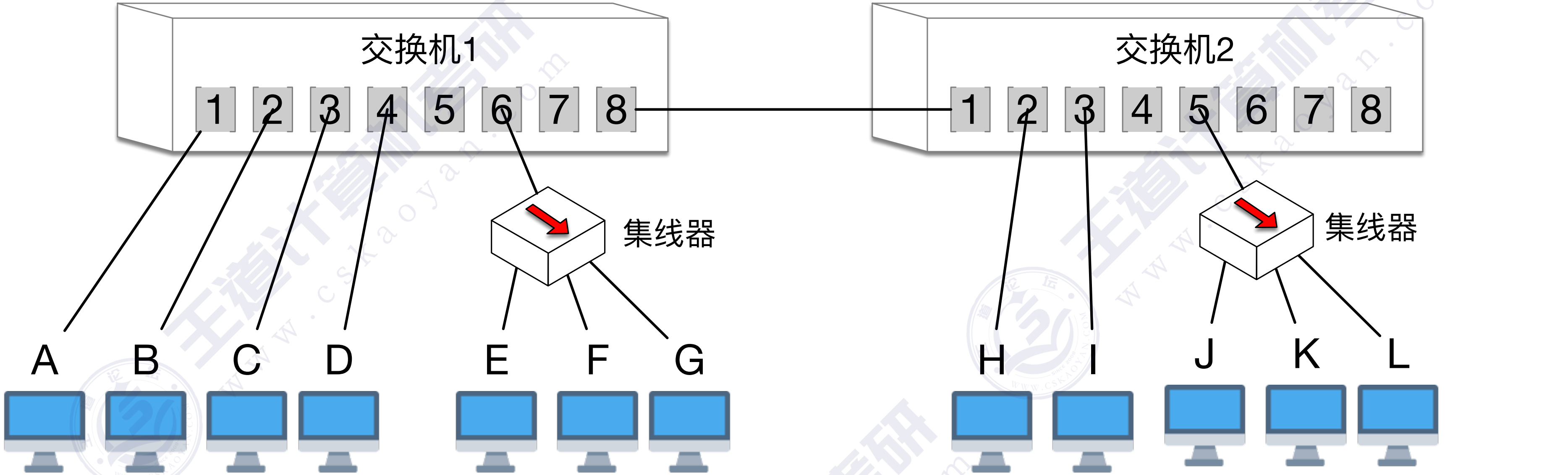
交换表1

MAC地址	端口号

交换表2

MAC地址	端口号

假设各节点知道彼此的MAC地址，并依次发送以下帧：
①A→C，②B→C，③C→A
④H→A，⑤H→C，⑥C→H
⑦J→广，⑧ K→J



假设MAC地址(48bit)
A = f8:ff:c2:6a:94:9a
B = f8:ff:c2:6a:94:9b
C = f8:ff:c2:6a:94:9c
D = f8:ff:c2:6a:94:9d
E = f8:ff:c2:6a:94:9e
F = f8:ff:c2:6a:94:9f
G = f8:ff:c2:6a:94:91
H = f8:ff:c2:6a:94:92
I = f8:ff:c2:6a:94:93
J = f8:ff:c2:6a:94:94
K = f8:ff:c2:6a:94:95
L = f8:ff:c2:6a:94:96

V2标准的以太网MAC帧：
6 6 2 N 4，收发协数验
CA

以太网交换机的自学习功能

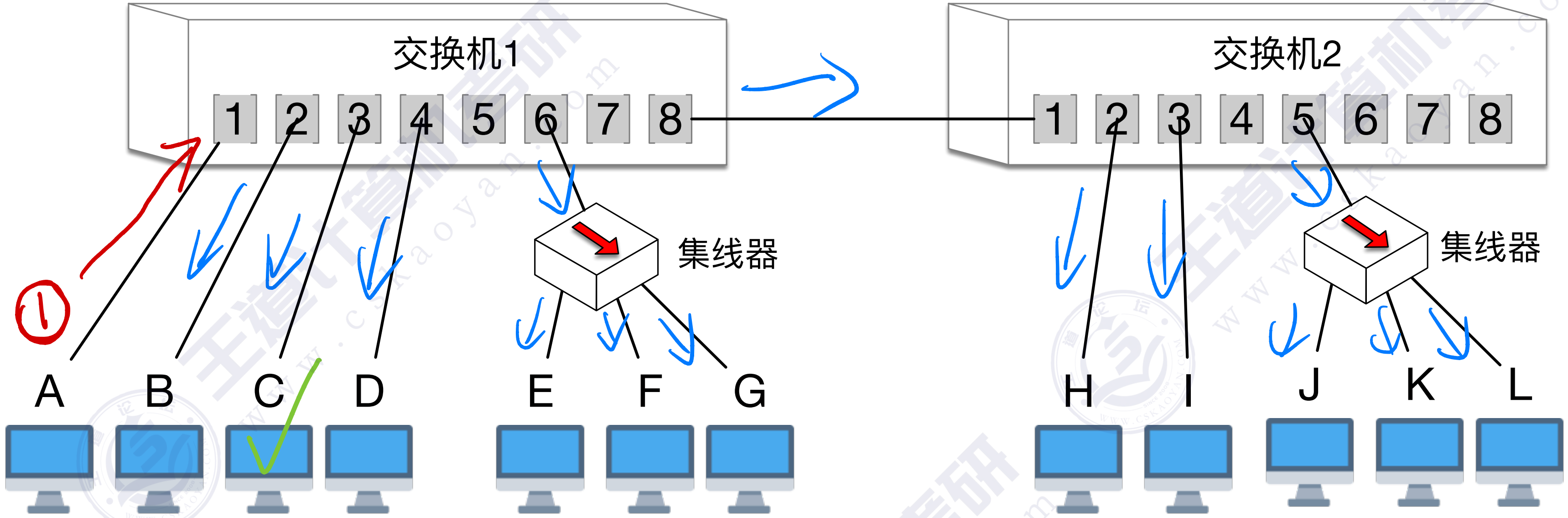
交换表1

MAC地址	端口号
(A) 9a	1

交换表2

MAC地址	端口号
(A) 9a	1

假设各节点知道彼此的MAC地址，并依次发送以下帧：
①A→C，②B→C，③C→A
④H→A，⑤H→C，⑥C→H
⑦J→广，⑧ K→J



假设MAC地址(48bit)
A = f8:ff:c2:6a:94:9a
B = f8:ff:c2:6a:94:9b
C = f8:ff:c2:6a:94:9c
D = f8:ff:c2:6a:94:9d
E = f8:ff:c2:6a:94:9e
F = f8:ff:c2:6a:94:9f
G = f8:ff:c2:6a:94:91
H = f8:ff:c2:6a:94:92
I = f8:ff:c2:6a:94:93
J = f8:ff:c2:6a:94:94
K = f8:ff:c2:6a:94:95
L = f8:ff:c2:6a:94:96

V2标准的以太网MAC帧：
6 6 2 N 4，收发协数验
C B

以太网交换机的自学习功能

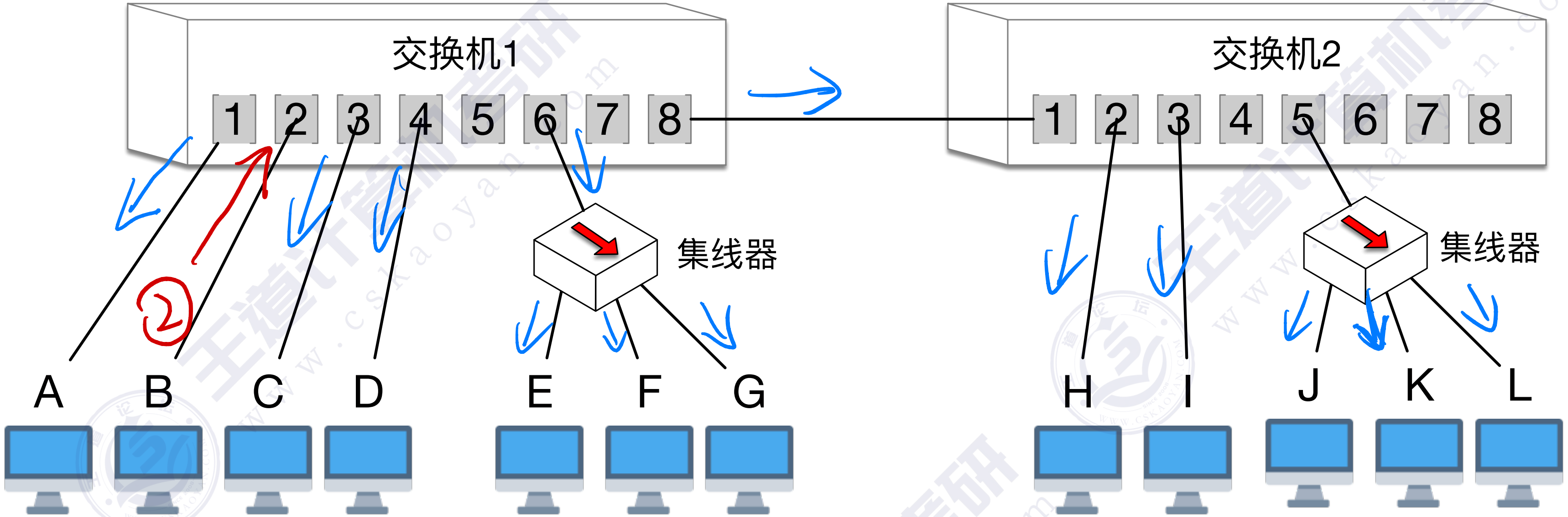
交换表1

MAC地址	端口号
(A) …. 9a	1
(B) …. 9b	2

交换表2

MAC地址	端口号
(A) …. 9a	1
(B) …. 9b	1

假设各节点知道彼此的MAC地址，并依次发送以下帧：
①A→C，②B→C，③C→A
④H→A，⑤H→C，⑥C→H
⑦J→广，⑧ K→J



假设MAC地址(48bit)
A = f8:ff:c2:6a:94:9a
B = f8:ff:c2:6a:94:9b
C = f8:ff:c2:6a:94:9c
D = f8:ff:c2:6a:94:9d
E = f8:ff:c2:6a:94:9e
F = f8:ff:c2:6a:94:9f
G = f8:ff:c2:6a:94:91
H = f8:ff:c2:6a:94:92
I = f8:ff:c2:6a:94:93
J = f8:ff:c2:6a:94:94
K = f8:ff:c2:6a:94:95
L = f8:ff:c2:6a:94:96

V2标准的以太网MAC帧：
6 6 2 N 4，收发协数验
A C

以太网交换机的自学习功能

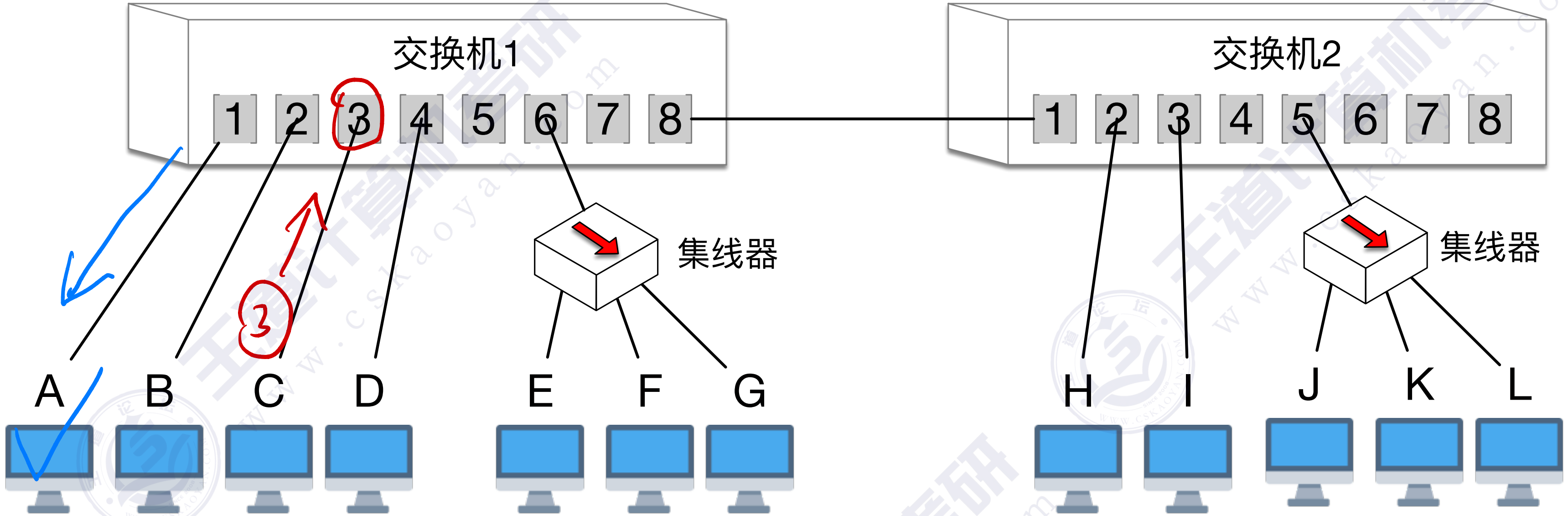
交换表1

MAC地址	端口号
(A) 9a	1
(B) 9b	2
(C) 9c	3

交换表2

MAC地址	端口号
(A) 9a	1
(B) 9b	1

假设各节点知道彼此的MAC地址，并依次发送以下帧：
①A→C，②B→C，③C→A
④H→A，⑤H→C，⑥C→H
⑦J→广，⑧ K→J



假设MAC地址(48bit)
A = f8:ff:c2:6a:94:9a
B = f8:ff:c2:6a:94:9b
C = f8:ff:c2:6a:94:9c
D = f8:ff:c2:6a:94:9d
E = f8:ff:c2:6a:94:9e
F = f8:ff:c2:6a:94:9f
G = f8:ff:c2:6a:94:91
H = f8:ff:c2:6a:94:92
I = f8:ff:c2:6a:94:93
J = f8:ff:c2:6a:94:94
K = f8:ff:c2:6a:94:95
L = f8:ff:c2:6a:94:96

以太网交换机的自学习功能

V2标准的以太网MAC帧：
6 6 2 N 4，收发协数验

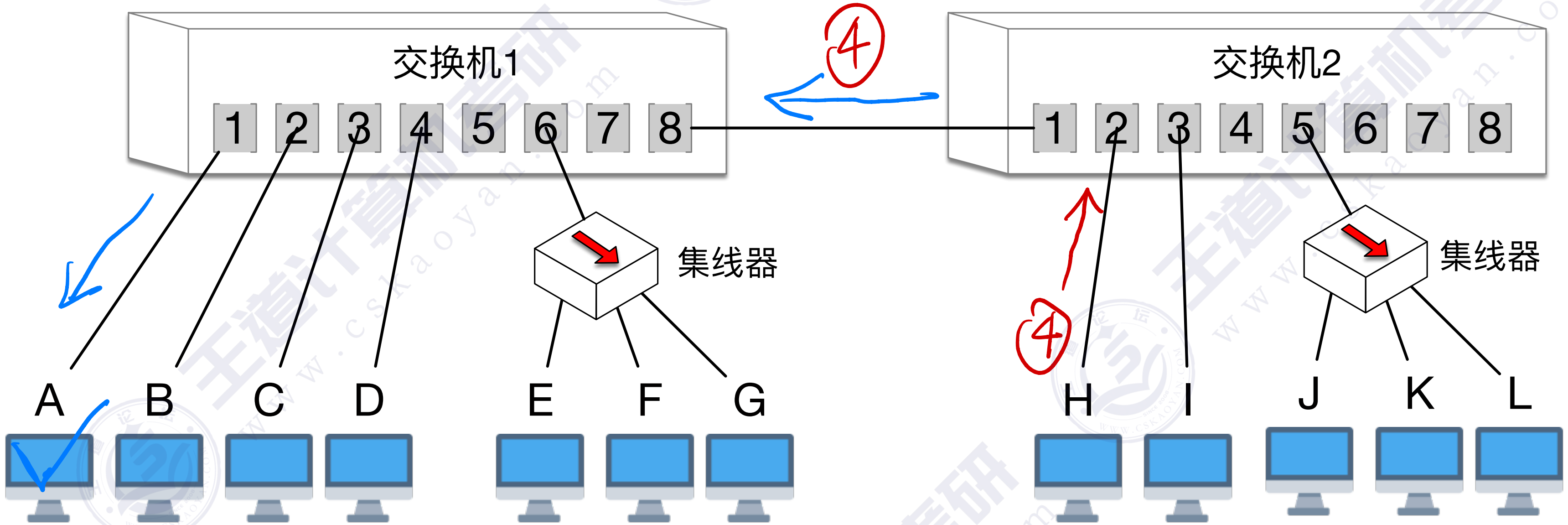
AH

交换表1

MAC地址	端口号
(A) 9a	1
(B) 9b	2
(C) 9c	3
(H) 92	8

交换表2

MAC地址	端口号
(A) 9a	1
(B) 9b	1
(H) 92	2



假设各节点知道彼此的MAC地址，并依次发送以下帧：
①A→C，②B→C，③C→A
④H→A，⑤H→C，⑥C→H
⑦J→广，⑧ **K→J**

假设MAC地址(48bit)
A = f8:ff:c2:6a:94:9a
B = f8:ff:c2:6a:94:9b
C = f8:ff:c2:6a:94:9c
D = f8:ff:c2:6a:94:9d
E = f8:ff:c2:6a:94:9e
F = f8:ff:c2:6a:94:9f
G = f8:ff:c2:6a:94:91
H = f8:ff:c2:6a:94:92
I = f8:ff:c2:6a:94:93
J = f8:ff:c2:6a:94:94
K = f8:ff:c2:6a:94:95
L = f8:ff:c2:6a:94:96

V2标准的以太网MAC帧：
6 6 2 N 4，收发协数验

C H

以太网交换机的自学习功能

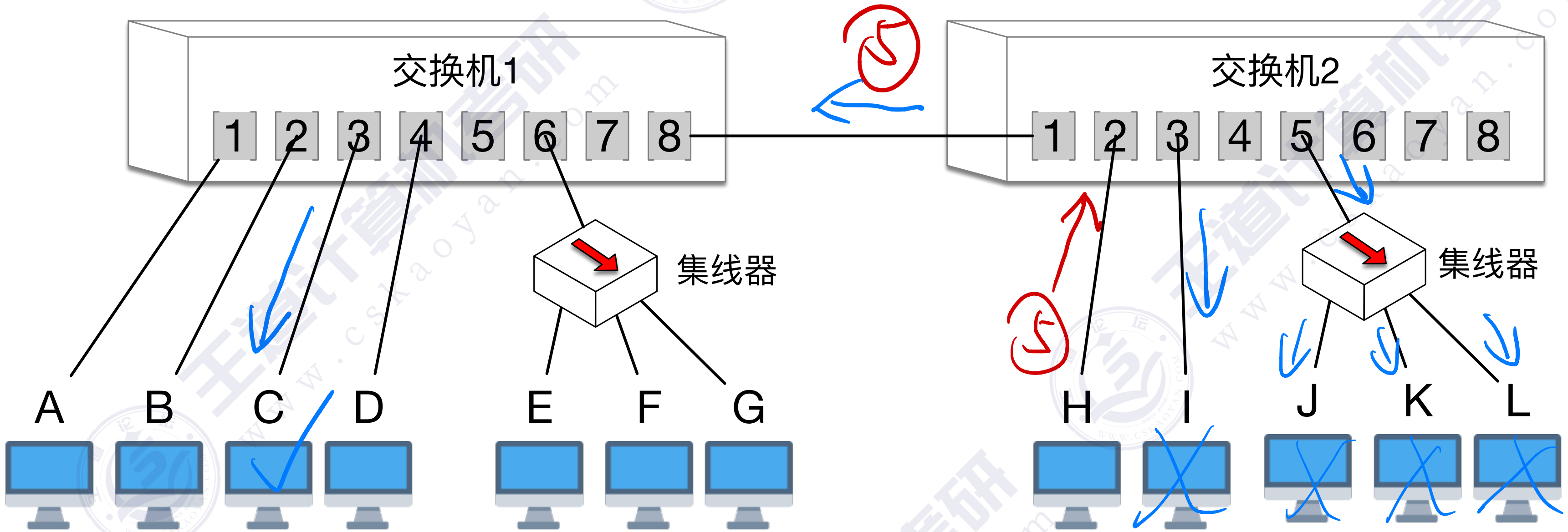
交换表1

MAC地址	端口号
(A) 9a	1
(B) 9b	2
(C) 9c	3
(H) 92	8

交换表2

MAC地址	端口号
(A) 9a	1
(B) 9b	1
(H) 92	2

假设各节点知道彼此的MAC地址，并依次发送以下帧：
①A→C，②B→C，③C→A
④H→A，⑤H→C，⑥C→H
⑦J→广，⑧ K→J



假设MAC地址(48bit)
A = f8:ff:c2:6a:94:9a
B = f8:ff:c2:6a:94:9b
C = f8:ff:c2:6a:94:9c
D = f8:ff:c2:6a:94:9d
E = f8:ff:c2:6a:94:9e
F = f8:ff:c2:6a:94:9f
G = f8:ff:c2:6a:94:91
H = f8:ff:c2:6a:94:92
I = f8:ff:c2:6a:94:93
J = f8:ff:c2:6a:94:94
K = f8:ff:c2:6a:94:95
L = f8:ff:c2:6a:94:96

V2标准的以太网MAC帧：

6 6 2 N 4，收发协数验

H C

以太网交换机的自学习功能

交换表1

MAC地址	端口号
(A) 9a	1
(B) 9b	2
(C) 9c	3
(H) 92	8

交换表2

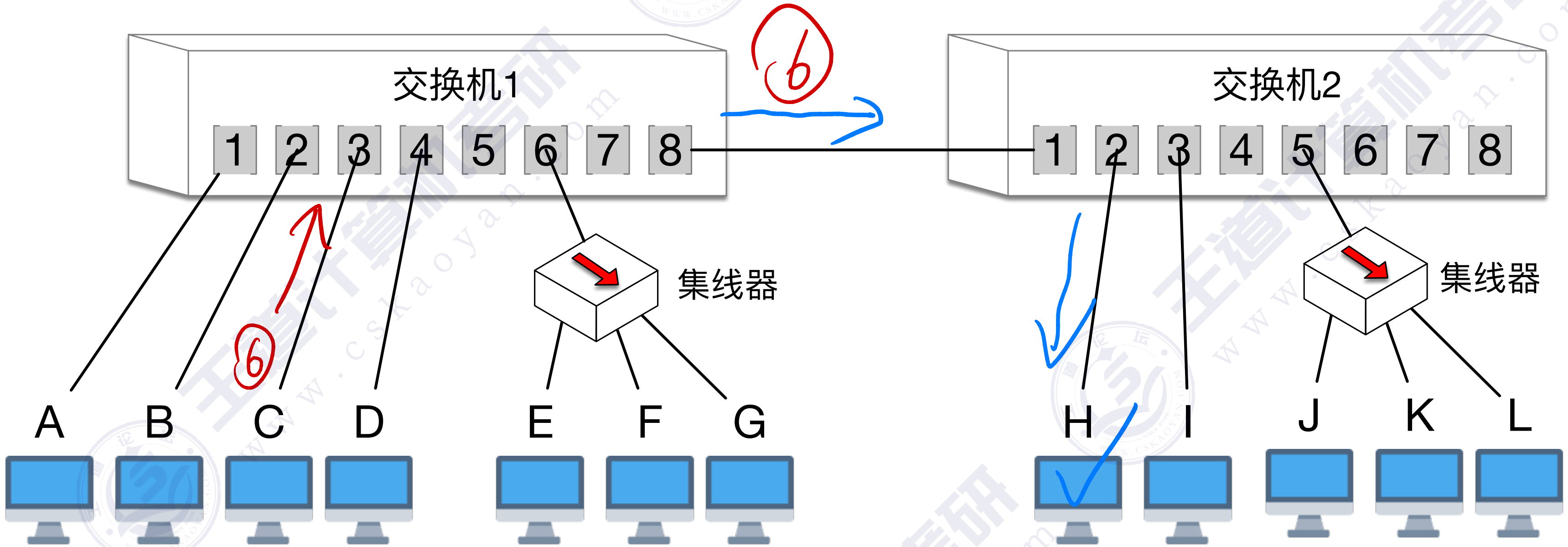
MAC地址	端口号
(A) 9a	1
(B) 9b	1
(H) 92	2
(C) 9c	1

假设各节点知道彼此的MAC地址，并依次发送以下帧：

①A→C，②B→C，③C→A

④H→A，⑤H→C，⑥C→H

⑦J→广，⑧ K→J



假设MAC地址(48bit)

A = f8:ff:c2:6a:94:9a

B = f8:ff:c2:6a:94:9b

C = f8:ff:c2:6a:94:9c

D = f8:ff:c2:6a:94:9d

E = f8:ff:c2:6a:94:9e

F = f8:ff:c2:6a:94:9f

G = f8:ff:c2:6a:94:91

H = f8:ff:c2:6a:94:92

I = f8:ff:c2:6a:94:93

J = f8:ff:c2:6a:94:94

K = f8:ff:c2:6a:94:95

L = f8:ff:c2:6a:94:96

V2标准的以太网MAC帧：

6 6 2 N 4，收发协数验

全1 J

以太网交换机的自学习功能

交换表1

MAC地址	端口号
(A) 9a	1
(B) 9b	2
(C) 9c	3
(H) 92	8
(J) 94	8

交换表2

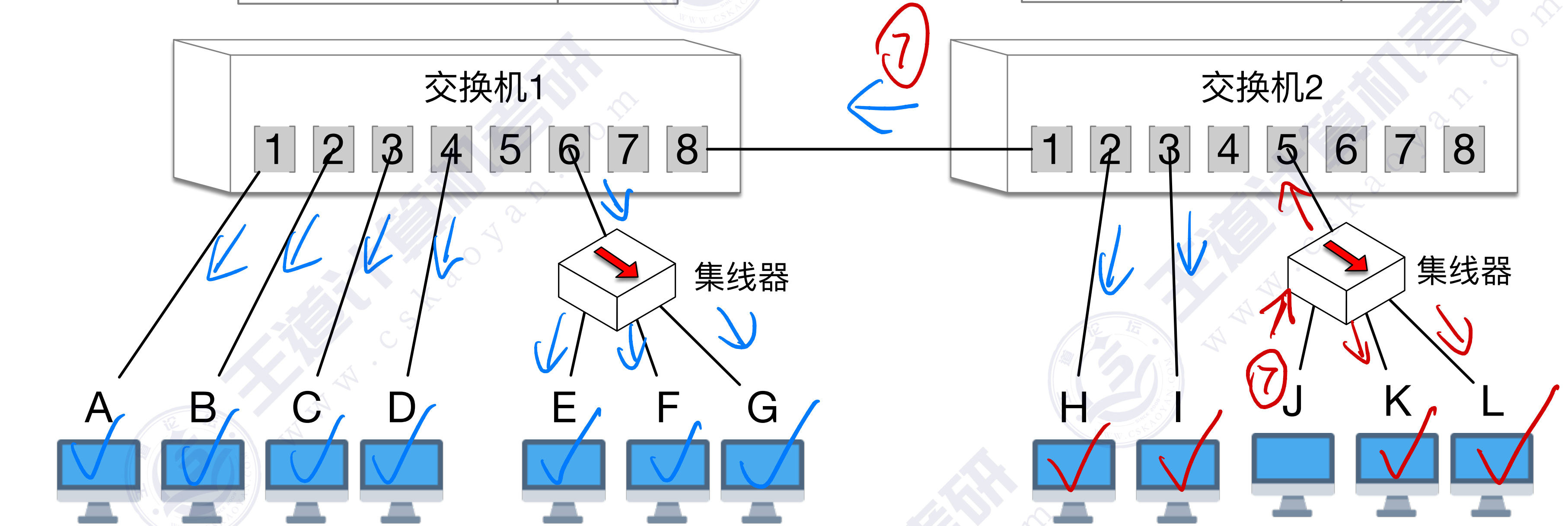
MAC地址	端口号
(A) 9a	1
(B) 9b	1
(H) 92	2
(C) 9c	1
(J) 94	5

假设各节点知道彼此的MAC地址，并依次发送以下帧：

①A→C，②B→C，③C→A

④H→A，⑤H→C，⑥C→H

⑦J→广，⑧ K→J



假设MAC地址(48bit)

A = f8:ff:c2:6a:94:9a

B = f8:ff:c2:6a:94:9b

C = f8:ff:c2:6a:94:9c

D = f8:ff:c2:6a:94:9d

E = f8:ff:c2:6a:94:9e

F = f8:ff:c2:6a:94:9f

G = f8:ff:c2:6a:94:91

H = f8:ff:c2:6a:94:92

I = f8:ff:c2:6a:94:93

J = f8:ff:c2:6a:94:94

K = f8:ff:c2:6a:94:95

L = f8:ff:c2:6a:94:96

以太网交换机的自学习功能

V2标准的以太网MAC帧：
6 6 2 N 4，收发协数验

J K

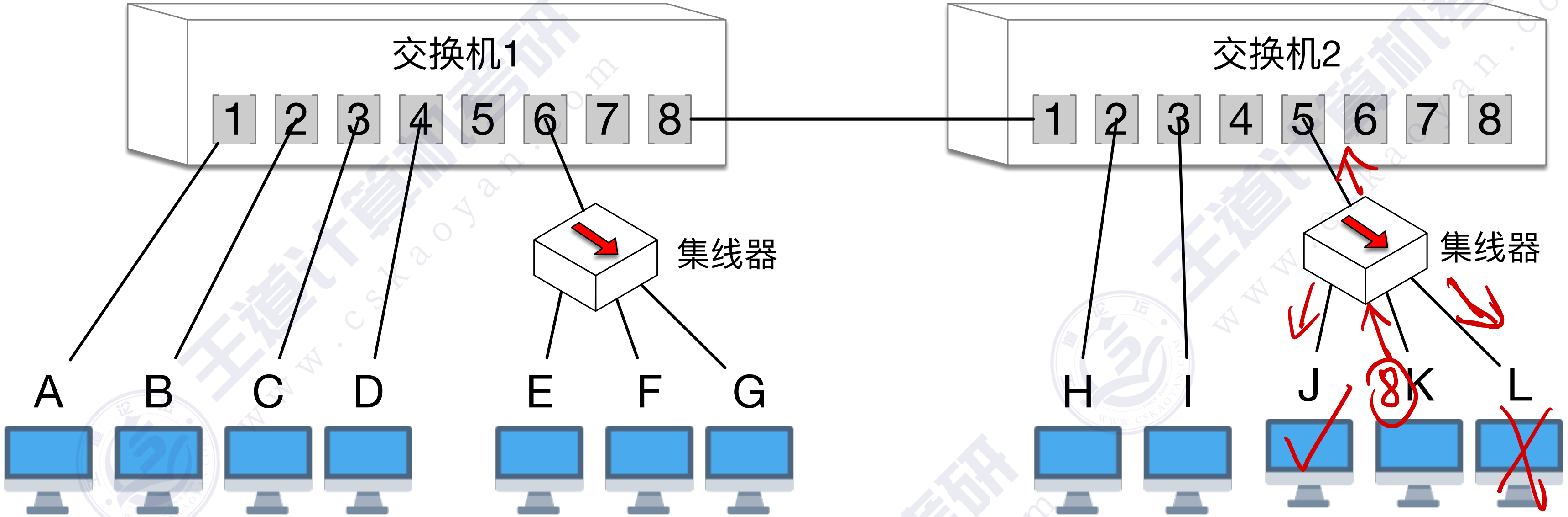
假设各节点知道彼此的MAC地址，并依次发送以下帧：
①A→C，②B→C，③C→A
④H→A，⑤H→C，⑥C→H
⑦J→广，⑧ K→J

交换表1

MAC地址	端口号
(A) 9a	1
(B) 9b	2
(C) 9c	3
(H) 92	8
(J) 94	8

交换表2

MAC地址	端口号
(A) 9a	1
(B) 9b	1
(H) 92	2
(C) 9c	1
(J) 94	5
(K) 95	5



假设MAC地址(48bit)
A = f8:ff:c2:6a:94:9a
B = f8:ff:c2:6a:94:9b
C = f8:ff:c2:6a:94:9c
D = f8:ff:c2:6a:94:9d
E = f8:ff:c2:6a:94:9e
F = f8:ff:c2:6a:94:9f
G = f8:ff:c2:6a:94:91
H = f8:ff:c2:6a:94:92
I = f8:ff:c2:6a:94:93
J = f8:ff:c2:6a:94:94
K = f8:ff:c2:6a:94:95
L = f8:ff:c2:6a:94:96

V2标准的以太网MAC帧：
6 6 2 N 4，收发协数验

以太网交换机的自学习功能

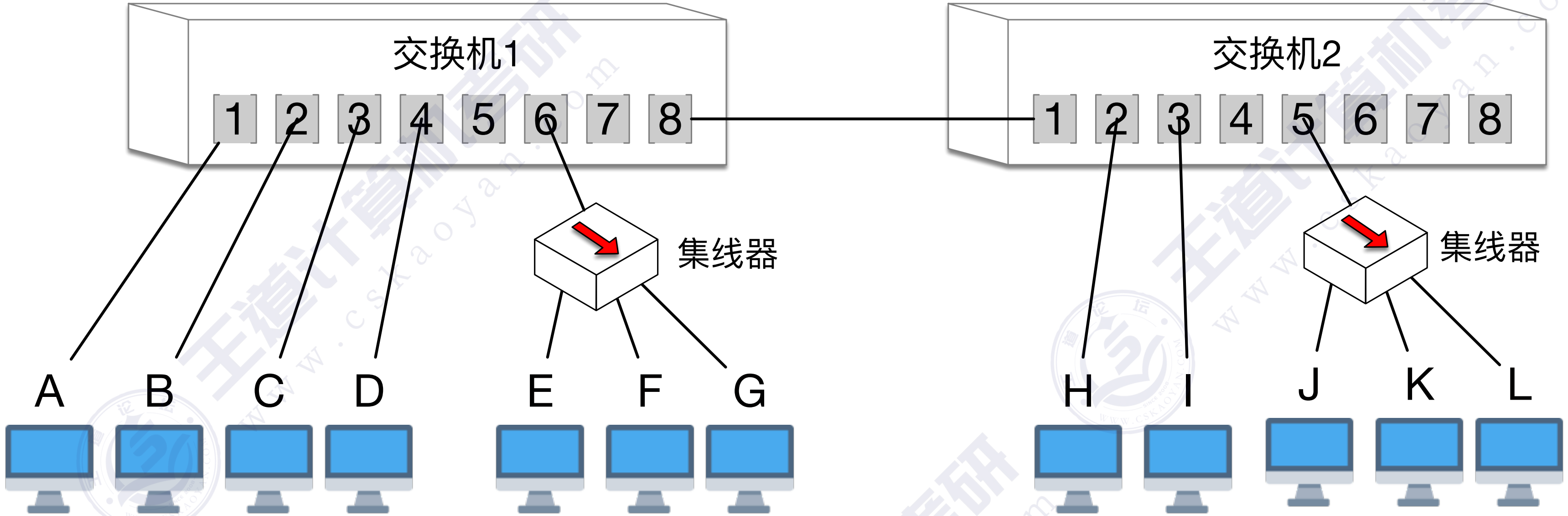
交换表1

MAC地址	端口号
(A)f8:ff:c2:6a:94:9a	1
(B)f8:ff:c2:6a:94:9b	2
(C)f8:ff:c2:6a:94:9c	3
(H)f8:ff:c2:6a:94:92	8
(J)f8:ff:c2:6a:94:94	8

交换表2

MAC地址	端口号
(A)f8:ff:c2:6a:94:9a	1
(B)f8:ff:c2:6a:94:9b	1
(H)f8:ff:c2:6a:94:92	2
(C)f8:ff:c2:6a:94:9c	1
(J)f8:ff:c2:6a:94:94	5
(K)f8:ff:c2:6a:94:95	5

假设各节点知道彼此的MAC地址，并依次发送以下帧：
①A→C，②B→C，③C→A
④H→A，⑤H→C，⑥C→H
⑦J→广，⑧K→J



假设MAC地址(48bit)
A = f8:ff:c2:6a:94:9a
B = f8:ff:c2:6a:94:9b
C = f8:ff:c2:6a:94:9c
D = f8:ff:c2:6a:94:9d
E = f8:ff:c2:6a:94:9e
F = f8:ff:c2:6a:94:9f
G = f8:ff:c2:6a:94:91
H = f8:ff:c2:6a:94:92
I = f8:ff:c2:6a:94:93
J = f8:ff:c2:6a:94:94
K = f8:ff:c2:6a:94:95
L = f8:ff:c2:6a:94:96

V2标准的以太网MAC帧：
6 6 2 N 4，收发协数验

以太网交换机的自学习功能

交换表1

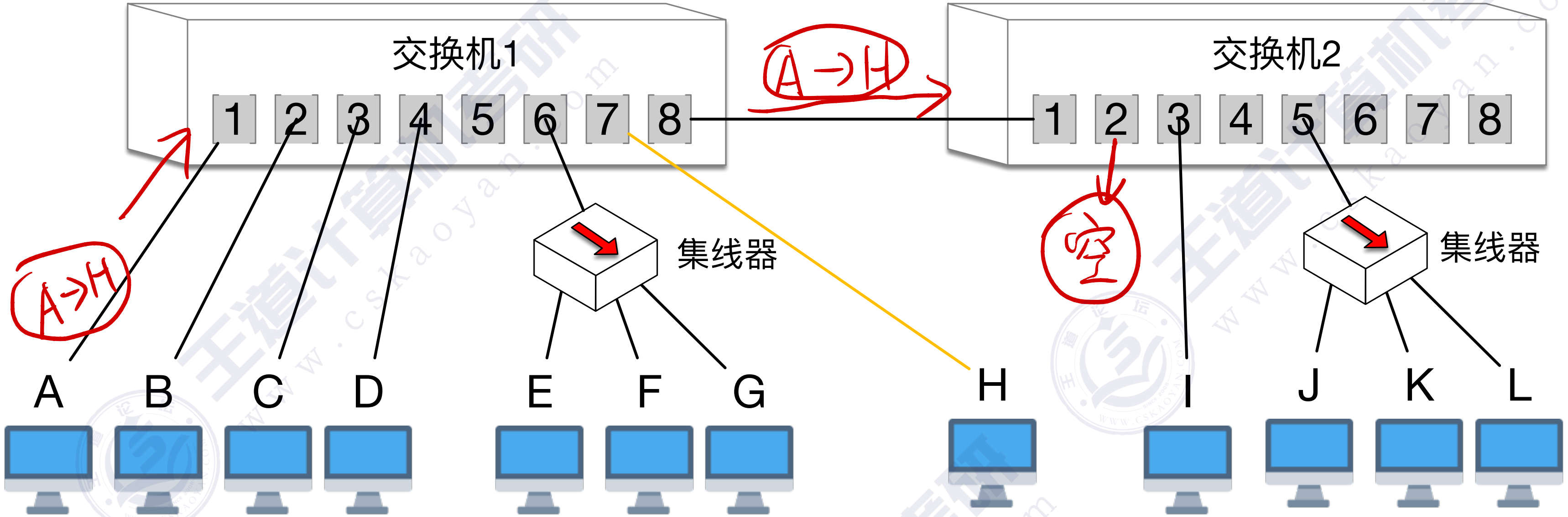
MAC地址	端口号
(A)f8:ff:c2:6a:94:9a	1
(B)f8:ff:c2:6a:94:9b	2
(C)f8:ff:c2:6a:94:9c	3
(H)f8:ff:c2:6a:94:92	8
(J)f8:ff:c2:6a:94:94	8

错

交换表2

MAC地址	端口号
(A)f8:ff:c2:6a:94:9a	1
(B)f8:ff:c2:6a:94:9b	1
(H)f8:ff:c2:6a:94:92	2
(C)f8:ff:c2:6a:94:9c	1
(J)f8:ff:c2:6a:94:94	5
(K)f8:ff:c2:6a:94:95	5

思考：
若H节点拔掉网线，换到交换机1-端口7，会发生什么？如何解决？



假设MAC地址(48bit)
A = f8:ff:c2:6a:94:9a
B = f8:ff:c2:6a:94:9b
C = f8:ff:c2:6a:94:9c
D = f8:ff:c2:6a:94:9d
E = f8:ff:c2:6a:94:9e
F = f8:ff:c2:6a:94:9f
G = f8:ff:c2:6a:94:91
H = f8:ff:c2:6a:94:92
I = f8:ff:c2:6a:94:93
J = f8:ff:c2:6a:94:94
K = f8:ff:c2:6a:94:95
L = f8:ff:c2:6a:94:96

V2标准的以太网MAC帧：
6 6 2 N 4，收发协数验

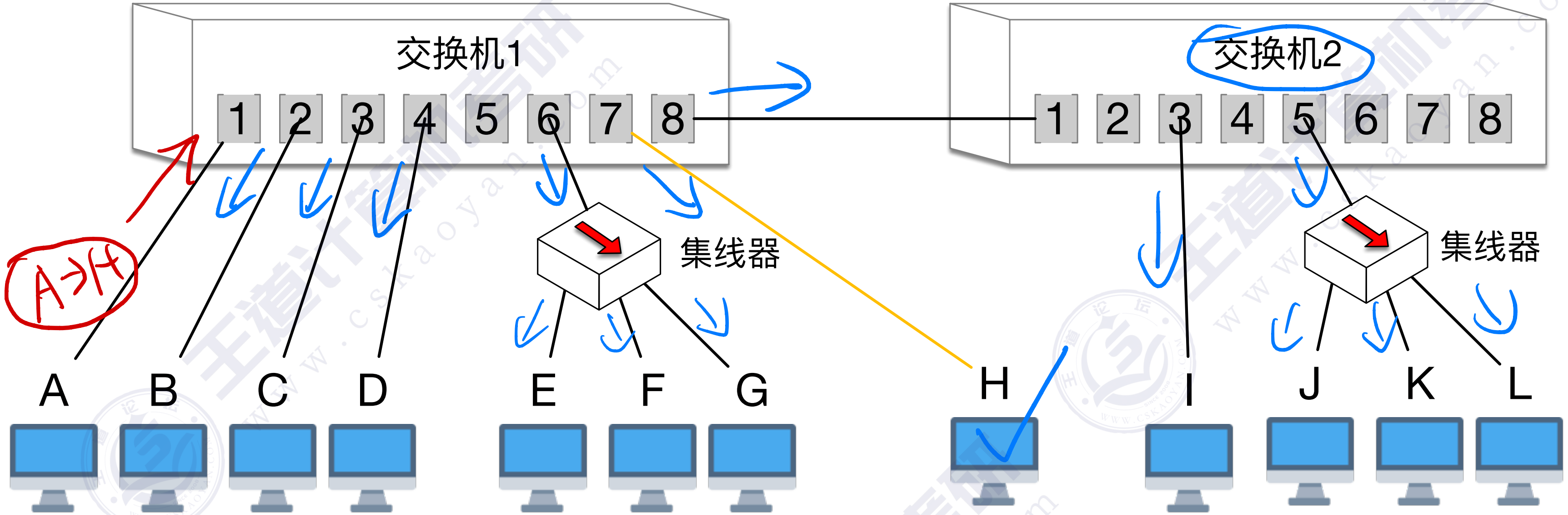
以太网交换机的自学习功能

交换表1

MAC地址	端口号
(A)f8:ff:c2:6a:94:9a	1
(B)f8:ff:c2:6a:94:9b	2
(C)f8:ff:c2:6a:94:9c	3
~~(H)f8:ff:c2:6a:94:92~~ 过期作废	~~8~~
(J)f8:ff:c2:6a:94:94	8

交换表2

MAC地址	端口号
(A)f8:ff:c2:6a:94:9a	1
(B)f8:ff:c2:6a:94:9b	1
~~(H)f8:ff:c2:6a:94:92~~ 过期作废	~~2~~
(C)f8:ff:c2:6a:94:9c	1
(J)f8:ff:c2:6a:94:94	5
(K)f8:ff:c2:6a:94:95	5



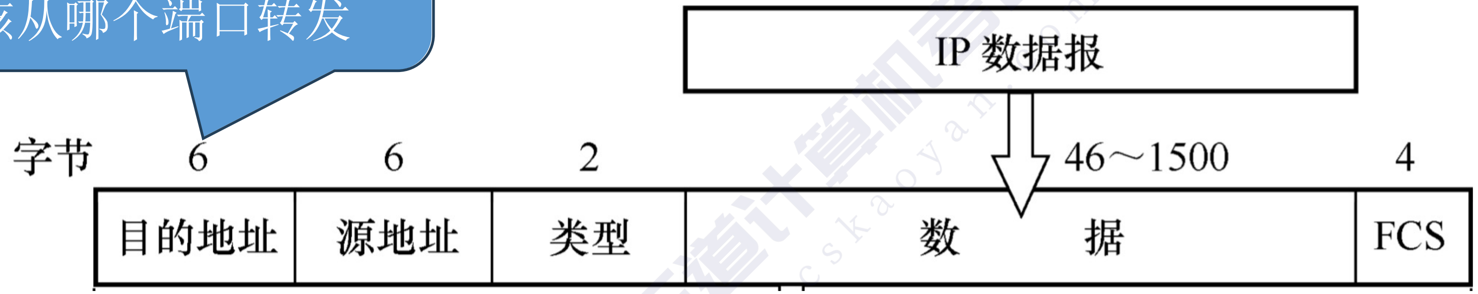
思考：
若H节点拔掉网线，换到交换机1-端口7，会发生什么？如何解决？

设置每个表项的有效期，过期自动作废

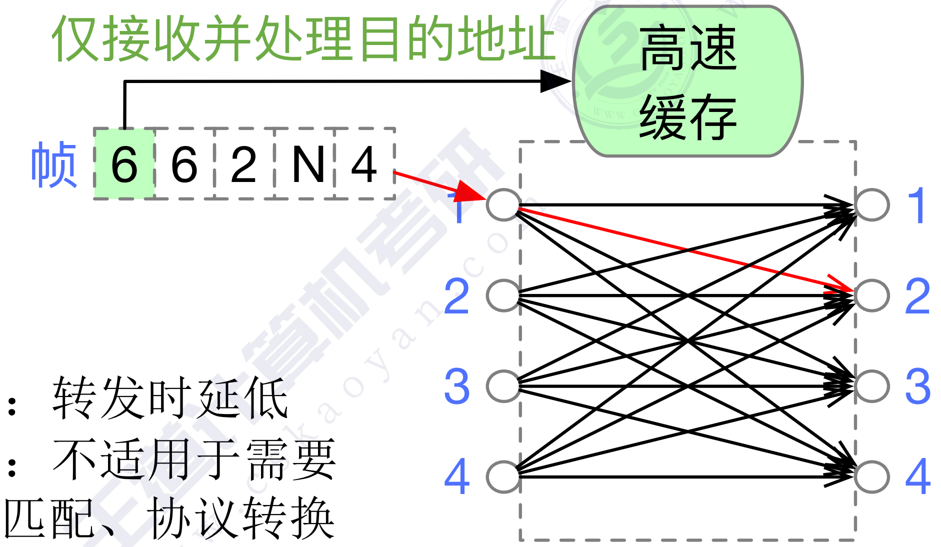
假设MAC地址(48bit)
A = f8:ff:c2:6a:94:9a
B = f8:ff:c2:6a:94:9b
C = f8:ff:c2:6a:94:9c
D = f8:ff:c2:6a:94:9d
E = f8:ff:c2:6a:94:9e
F = f8:ff:c2:6a:94:9f
G = f8:ff:c2:6a:94:91
H = f8:ff:c2:6a:94:92
I = f8:ff:c2:6a:94:93
J = f8:ff:c2:6a:94:94
K = f8:ff:c2:6a:94:95
L = f8:ff:c2:6a:94:96

交换机：直通交换 vs 存储转发交换

以太网交换机只需要知道帧的目的地址，就能决定应该从哪个端口转发

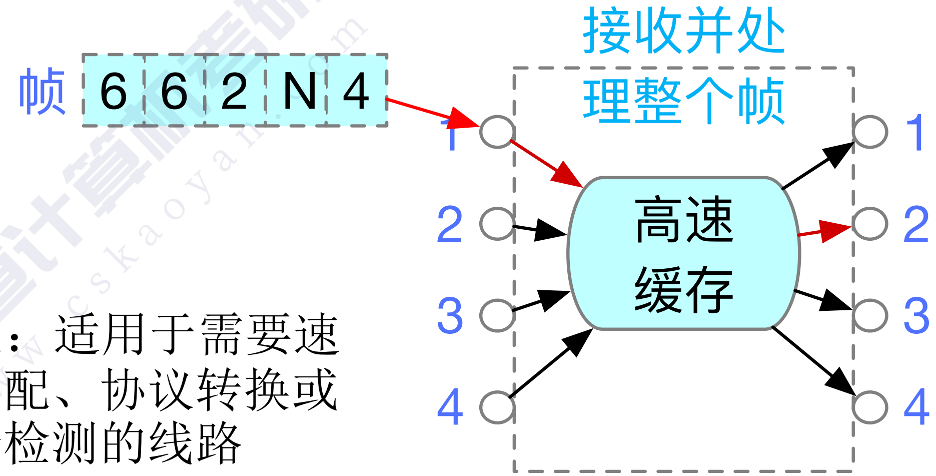


V2标准的以太网MAC帧：
6 6 2 N 4，收发协数验



直通式交换方式

优点： 转发时延低
缺点： 不适用于需要速率匹配、协议转换或差错检测的线路



存储转发交换方式

优点： 适用于需要速率匹配、协议转换或差错检测的线路
缺点： 转发时延高

例题：2013年真题38

38．对于 100Mbps 的以太网交换机，当输出端口无排队，以直通交换（cut-through switching）方式转发一个以太网帧（不包括前导码）时，引入的转发延迟至少是（　　）。

A．0μs　　B．0.48μs　　C．5.12μs　　D．121.44μs

直通交换：交换机仅需接收并处理目的地址（6字节，48bit）

接收 48bit 所需时间至少 = 48b/100Mbps = 0.48 μs